

Cracovian

Theoretical background

Cracovians are alternative form of defining complexes and algebra between them. Biggest difference between matrix and cracovian is the definition of multiplication, which for Cracovians, is more similar to polynomial multiplication. Theory was crafted by Tadeusz Banachiewicz and name comes from our beautiful city of Cracow.

Tadeusz Banachiewicz - a brilliant astronomer, mathematician and geodesist – created that theory to answer the need of better tools for solving specific problems in astronomy and geodesy. One such problem is pinpointing correct orbit coordinates of multiple astronomic bodies.

Special Cracovians, such as the ones with the same number of columns and rows, are called quadratic (kwadraste). Quadratic Cracovians with all other elements equal to 0 except for the diagonal elements (a_{ij}), are called diagonal (przekątniowe). Special case of diagonal Cracovian where all the values are equal to 1 ($a_{ij} == 1$) are called τ (tau from Greek “tautos” meaning “the same”).

Cracovian addition and subtractions are defined like those of regular matrices.

Cracovian multiplication is defined similarly to polynomial multiplication, meaning each column of row of first cracovian must be multiplied by each element of the second cracovian. Multiplication can only happen when both cracovians have the same number of rows.

$$\begin{pmatrix} a_{1,1} & \cdots & a_{p,1} \\ a_{1,2} & \cdots & a_{p,2} \\ \vdots & \ddots & \vdots \\ a_{1,m} & \cdots & a_{p,m} \end{pmatrix} \begin{pmatrix} b_{1,1} & \cdots & b_{n,1} \\ b_{1,2} & \cdots & b_{n,2} \\ \vdots & \ddots & \vdots \\ b_{1,m} & \cdots & b_{n,m} \end{pmatrix} = \begin{pmatrix} c_{1,1} & \cdots & c_{p,1} \\ c_{1,2} & \cdots & c_{p,2} \\ \vdots & \ddots & \vdots \\ c_{1,n} & \cdots & c_{p,n} \end{pmatrix}$$

Where $c_{l,k} = a_{l,1}b_{k,1} + a_{l,2}b_{k,2} + \cdots + a_{l,m}b_{k,m}$

For example:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae + cg & be + dg \\ af + ch & bf + dh \end{pmatrix}$$

$$\begin{pmatrix} 3 & -2 & 5 \\ -1 & 0 & 4 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -4 \\ -2 & 3 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} 5 & -2 & -3 \\ -9 & 2 & -2 \end{pmatrix}$$

Equation 1 - Example of 3x3 and 3x2 cracovian multiplication.

$$\begin{array}{cc}
 \begin{array}{c} 0 \\ \left\{ \begin{array}{cc} 3 & -2 \\ 1 & -1 \end{array} \right\} \end{array} & \begin{array}{c} 1 \\ \left\{ \begin{array}{cc} 1 & -4 \\ -2 & 3 \end{array} \right\} \end{array} \\
 \begin{array}{c} 0 \\ \left\{ \begin{array}{cc} 1 & -2 \\ -9 & 5 \end{array} \right\} \end{array} & \begin{array}{c} 1 \\ \left\{ \begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right\} \end{array}
 \end{array}
 =
 \begin{array}{cc}
 \begin{array}{c} 0 \\ \left\{ \begin{array}{cc} 1 & -2 \\ -9 & 5 \end{array} \right\} \end{array} & \begin{array}{c} 1 \\ \left\{ \begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array} \right\} \end{array}
 \end{array}$$

Handwritten annotations on the result matrices:
 Top-left matrix: $3-2$ above the 1, $-12+3$ below the -9.
 Top-right matrix: $-2+2$ above the 0, $8-3$ below the 5.

Equation 2 - Example of 2x2 and 2x2 cracovian multiplication

Goal

An ancient rotatory calculator machine has broken! It was vital for solar system interplanetary trajectory calculations and all the traffic was using it. We cannot go to Mars without it. We need your help to fix it! One broken part seems to be the cracovian multiplier.

You need to provide a robust Cracovian multiplier with configurable cracovian sizes and fixed-point arithmetic. Input and output Cracovians are to be sent as a 2x2 subcracovians with 4 mantissa values and 1 common exponent.

Configuration is sent as a first sample on the data path in this format:

$CRAC = \{ num_pair, num_rows, num_col_a, num_col_b \}$ where:

num_pair = 8 - bit unsigned number of pairs of cracovian for multiplication

num_rows = 8 - bit unsigned number of rows of cracovian factors

num_col_a, num_col_b = 8 - bit unsigned each, number of columns for factor a and factor b cracovians.

Configuration signal is provided together with i_first signal being asserted to 1'b1.

There might be multiple types of cracovian pairs needed for multiplication but all of them can be decomposed into 2x2 subcracovians.

2x2 subcracovian format is as follows:

$CRCV_2X2 = \{ mantisa[1][1], mantisa[1][0], mantisa[0][1], mantisa[0][0], exp \}$

Where $MANT_W = 7$ bits and $EXP_W = 4$ bits

And where $mantisa[0][1]$ points to mantisa value at row [0] and column [1].

Example of sample composition for 6x4 and 6x6 cracovians.

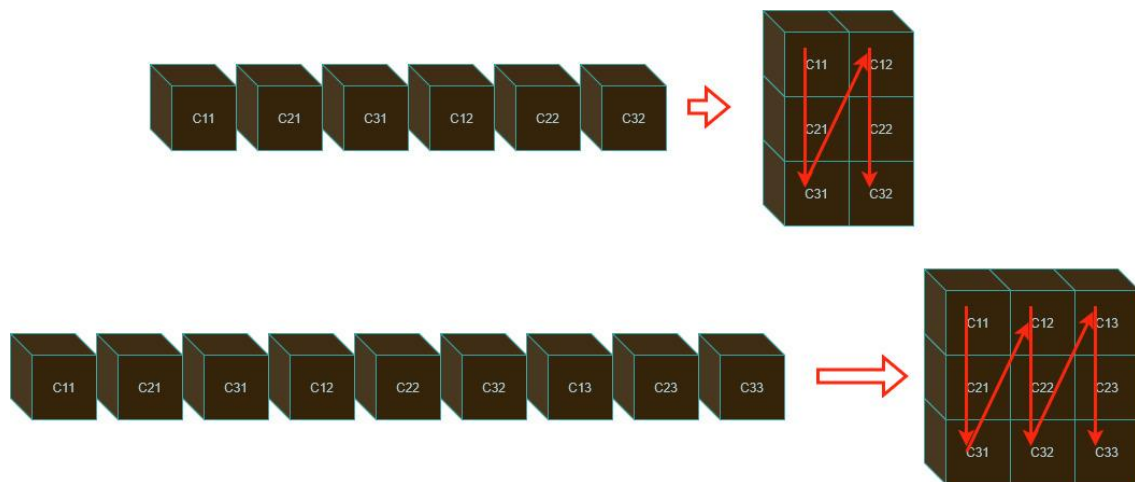


Figure 1 - Subcracovian sample decomposition schematic.

Input cracovians will arrive in series with *num_pair* number of pairs. Total number of samples has to be calculated by YOU 🙌.

Cracovians appear on the bus one after another with factor a being first and factor b second. Then the next pair:

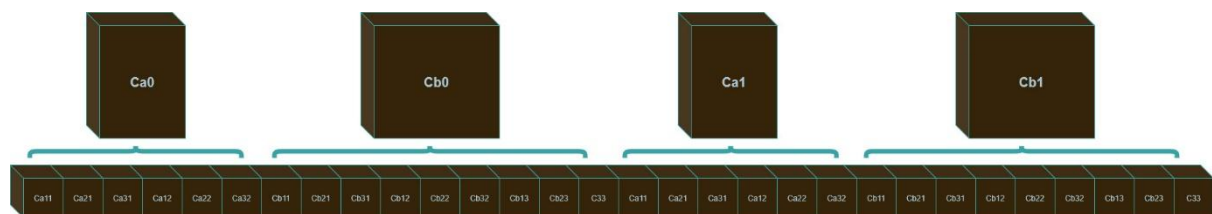


Figure 2 - Example input sequence with 2 pairs of cracovians

Ca0 – first factor of the first pair, Cb0 - second factor of the first pair

Ca1 – first factor of the second pair, Cb1 - second factor of the second pair

Provided module

You can edit only the *task_9.sv* file. If you wish, you may add other files for supportive submodules.

The module has the following inputs and outputs:

Table 1 - List of inputs and outputs of the *task_9* module.

Signal name	Signal type	Bit length
i_clk	Input	1 bit
i_valid	Input	1 bit
i_rst	Input	1 bit
i_data	Input	32 bits
i_last	Input	1 bit

i_first	Input	1 bit
o_data	Output	32 bits
o_valid	Output	1 bit
o_last	Output	1 bit
o_first	Output	1 bit

The reset is synchronous and active high.

Input and output interfaces

In one packet task receives configuration signal and a number of input data samples specified by the configuration. Each data sample is 32 bits. The last sample in a packet is indicated by signal *i_last*. The task should send back configuration signal on first sample and then the results cracovian multiplications as output samples, 32 bits each, and should indicate the last sample in the output packet by providing signal *o_last*.

Task receives data in the way shown in the figure below:

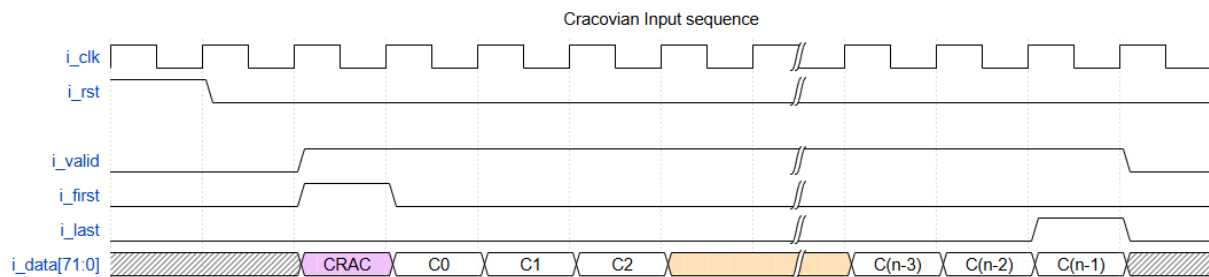


Figure 3 - Input waveforms

To send data from the task in a proper way one needs to provide the configuration signal for the output together with *o_first* flag, the *o_valid* signal that is synchronized with data on the *o_data* bus and *o_last* signal that is synchronized with the last sample in output packet.

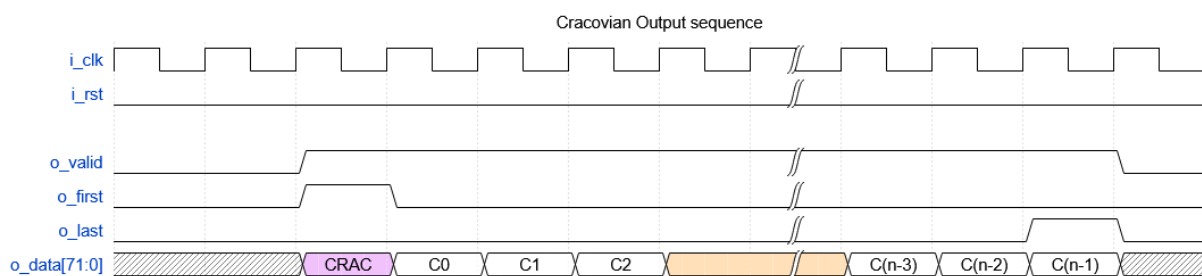


Figure 4 - Output waveforms

Output data can be sent as a continuous stream, or it can have regular or irregular gaps. The only requirement is providing *o_valid* and *o_last* signals.

Evaluating the task

Task is evaluated by reading the output and matching it to the software model. First sample of output (CRAC) must contain correct sizes and number of calculated cracovians.

$CRAC = \{ num_crac, num_rows, '0, num_col \}$

Where similarly to input *CRAC*:

num_crac = 8 - bit unsigned number of output cracovians.

num_row = 8 - bit unsigned number of rows (must be multiple of 2).

num_col = 8 - bit unsigned number of columns (must be multiple of 2).

Biggest cracovian multiplication cases are $64 \times 64 * 64 \times 64$ and maximum total number of input samples is 2048.

The task can be awarded maximum of 20 points (15 base and 5 bonus).

In Development mode, 3. In Evaluation mode 15 points (3 for each golden vector).

For more information, check chapter 2.3.1 **Testing modes** in the **FPGA_Hackathon_2026_Project_Guide** document. Only evaluation mode is considered for final score.

Additional points may be awarded for resource utilization while the latency does not extend 330 μ s. For more information, check chapter 2.3.2 **Bonus Points** in the **FPGA_Hackathon_2026_Project_Guide** document).